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LearnPath AI

A multi-agent system for personalized learning and knowledge-gap remediation

Proof of concept for Foundations of Agentic AI



Planner · Diagnostic · Tutor · Critic

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The problem: gaps hide behind scores

Problem statement

A learner who struggles with backpropagation is usually weak in something underneath it, such as the chain rule or gradient descent. The quiz score only says "weak in backpropagation", so the learner does not know what to fix first or how long it will take.

Why personalized learning suits agents

It is a loop (diagnose, plan, teach, assess, adapt) over a prerequisite graph. Plans and quizzes can be checked against the data, and it needs a human in the loop.



Symptoms, not causes

A low score hides the real gap underneath it.



One pace for everyone

Tutors cannot diagnose every learner at scale.



Plans nobody checks

Nothing tests whether the time or the quiz is right.

From one-size-fits-all to a personal path

Today

1. Same syllabus and pace for everyone
2. A score per topic, with no cause
3. The learner guesses what to study first
4. Failing a quiz means repeating the same material

LearnPath AI

1. Path built from the learner's own gaps and time
2. Root-cause analysis on the prerequisite graph
3. Prerequisites first, with hours per topic and an honest feasibility check
4. Re-teach in a new way, then a human tutor if still stuck

Finds the root cause, not the symptom

Every plan and quiz is validated

Adapts, and hands over to a human

Four agents, one orchestrator



Planner

Understands the goal

- Parses topics and time budget (rules or LLM)
- Refuses off-topic requests and other learners' data
- Writes the task plan



Diagnostic

Finds the gaps

- Builds the mastery map from quiz history
- Walks the prerequisite graph
- Separates root causes from downstream gaps



Tutor

Plans and teaches

- Builds a time-boxed study path
- Explains topics and prepares quizzes
- Re-teaches after a failed quiz



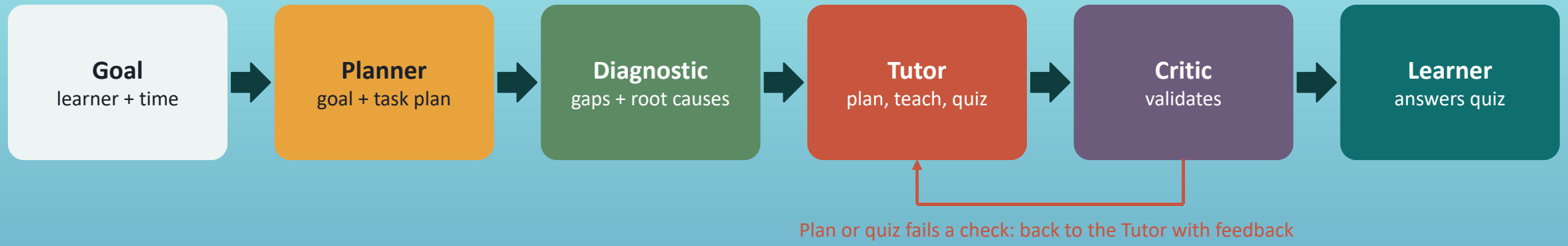
Critic / Assessor

Validates and decides

- Re-derives the facts to check plans and quizzes
- Grades answers and updates mastery
- Decides: cleared, re-teach or escalate

Orchestrator (LangGraph): pauses for the learner's answers (human in the loop), retries a failed plan or quiz, and escalates to a human tutor.

Orchestration: diagnose, teach, validate, adapt



1 Validate the plan

8 checks against recomputed data. Fail: the Tutor rebuilds it (max 2 attempts), then a human tutor is asked.

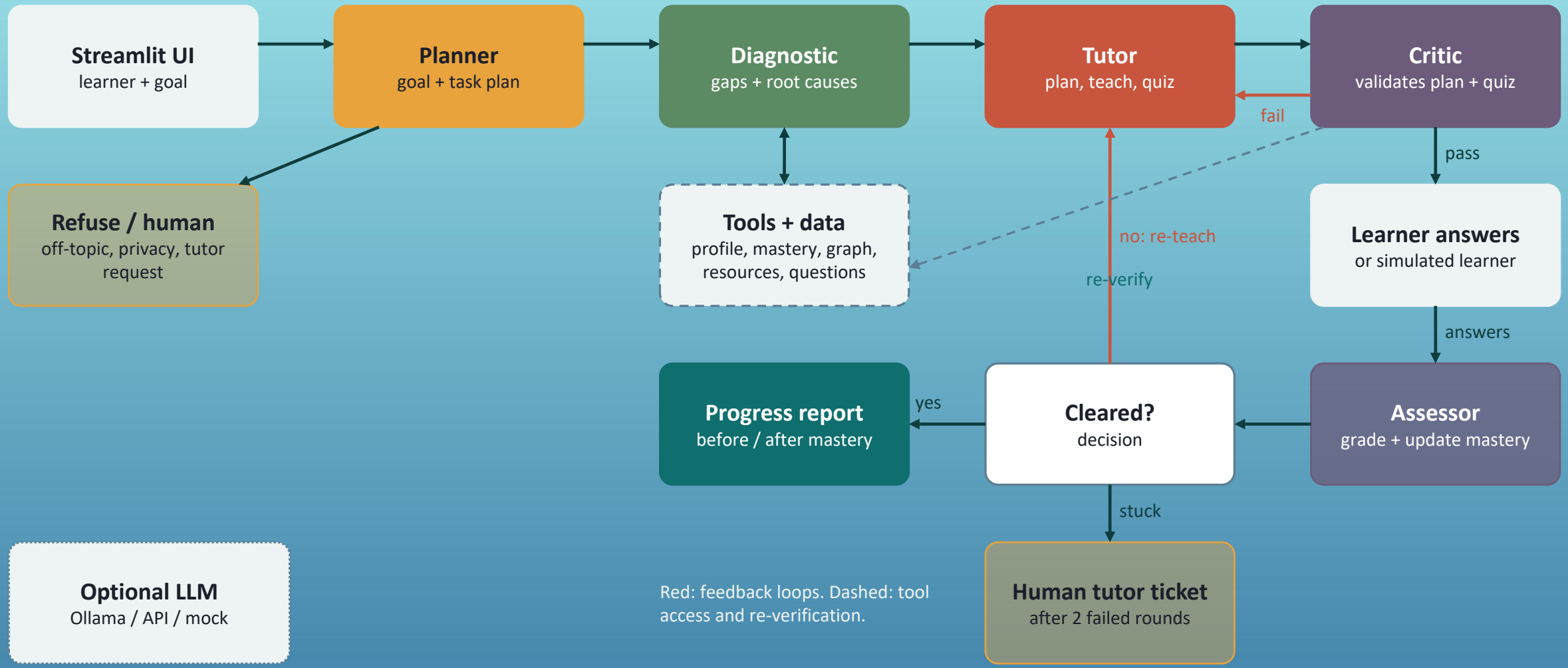
2 Validate each quiz

Format, topic, difficulty, no duplicates, explanations, freshness. LLM-written questions get an independent self-check.

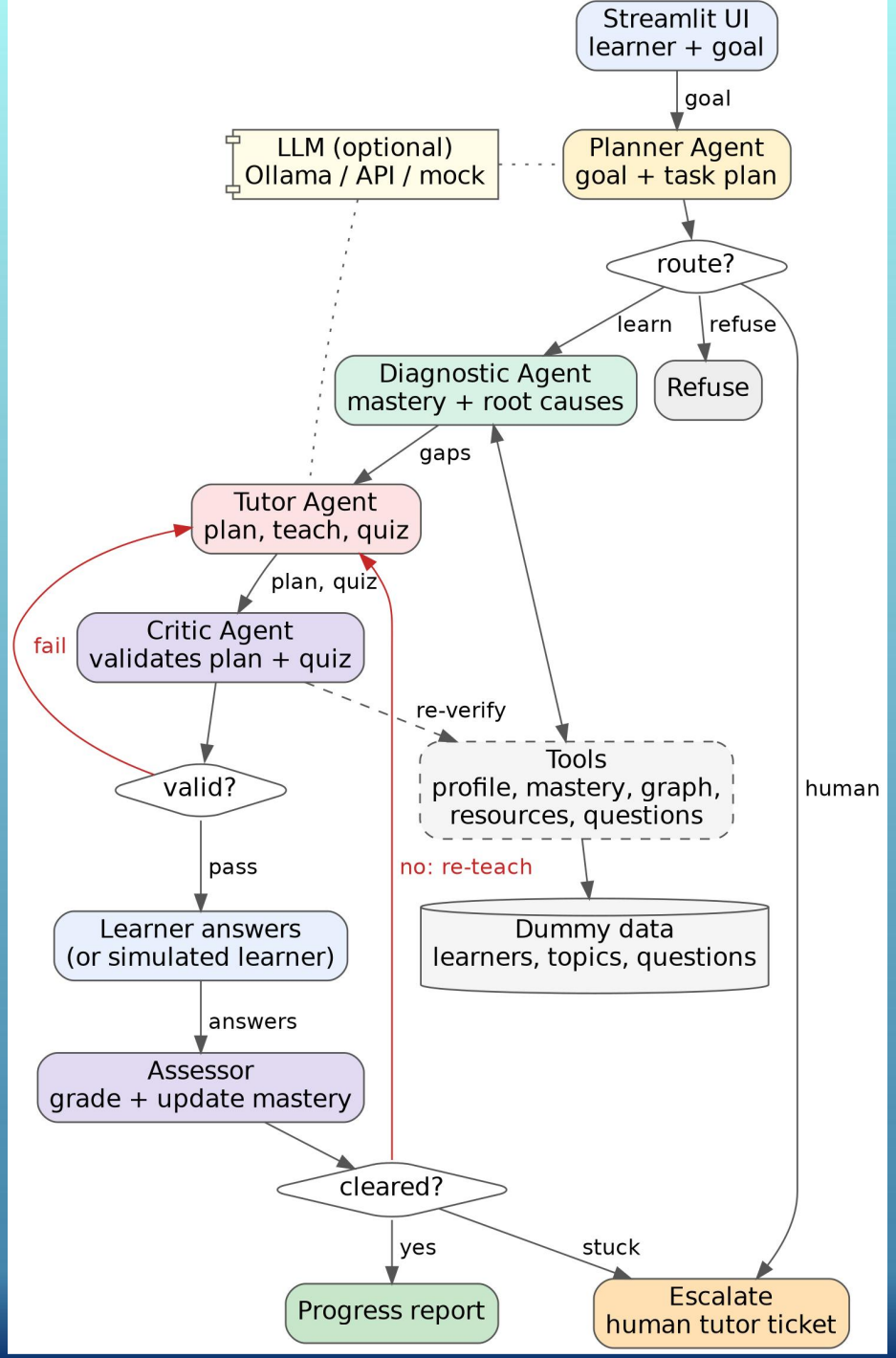
3 Assess and adapt

Cleared: next topic. Not cleared: re-teach with a new explanation. Still not cleared after 2 rounds: human tutor.

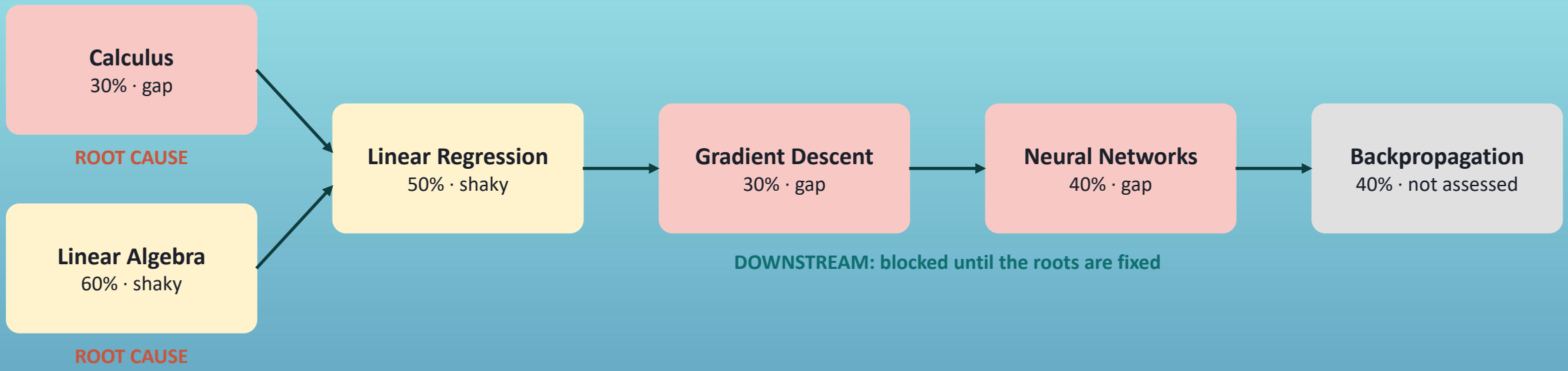
System architecture



System architecture



Root-cause diagnosis on the prerequisite graph



Learner L001 whose goal is Backpropagation. Some prerequisite edges are omitted for clarity.

Root cause

A topic that needs work (mastery under 70%) and whose prerequisites do not. Fix these first.

Hours needed

Estimated hours x (1 - mastery), rounded to half hours. Here 10.5 h of a 15 h budget.

Order and honesty

Prerequisites first, weakest first. If time is short, the plan says so and states the days really needed.

Optional LLM: Ollama, an API, or none

Agent	What the LLM may do	If it fails or is off
Planner	Parses a free-text goal into topics and a time budget	Rule-based parser
Tutor	Writes the explanation for each topic and re-teach round	Template from topic notes
Tutor	Writes extra practice questions (off by default)	Curated question bank
Critic	Answers an LLM-written question to verify its answer key	Question is rejected

Ollama setup: install Ollama, run ollama pull llama3.1, pip install langchain-ollama, then pick it in the sidebar and press Check connection.

Works with no LLM

Mock mode is the default: no key, no download, no internet.

Free and local

Ollama runs a model on your own machine.

Rules keep control

Guardrails stay in code; every LLM call is counted and has a fallback.

Tools, data and technology

Tool	Used by	Returns
get_learner_profile	Diagnostic	Name, level, quiz history size
compute_mastery_map	Diagnostic, Critic	Mastery and status per topic
get_prerequisite_closure	Diagnostic	Goal topics plus prerequisites
find_resources	Tutor	Videos, articles, exercises
get_practice_questions	Tutor	Questions by difficulty, unseen first
escalate_to_tutor	Orchestrator	Opens a human tutor ticket

Stack: Python, Streamlit, LangGraph (with an interrupt for the learner's answers), LangChain tools, Pydantic. Optional: Ollama or an API model.

12 topics in a prerequisite graph

60 questions, 5 per topic

4 learners with quiz history

Demo: the Critic catches a faulty plan

1. **Planner:** goal Backpropagation, 10 days x 1.5 h = 15 h
2. **Diagnostic:** root causes: Calculus and Linear Algebra
3. **Tutor:** builds the plan (corrupted on purpose)
4. **Critic:** plan FAILED, 2 of 8 checks passed
5. **Tutor:** rebuilds it with the feedback: 6 topics, 10.5 of 15 h
6. **Critic:** plan PASSED, 8 of 8 checks
7. **Critic:** first quiz also corrupted and rejected; valid quiz on retry

What the Critic found in the first plan

- Total hours: plan says 12.0 h, recomputed 10.5 h
- Backpropagation scheduled before its prerequisites
- Time spent on Python, already mastered
- Resource RES-FAKE-9 does not exist
- Over the daily limit on day 1

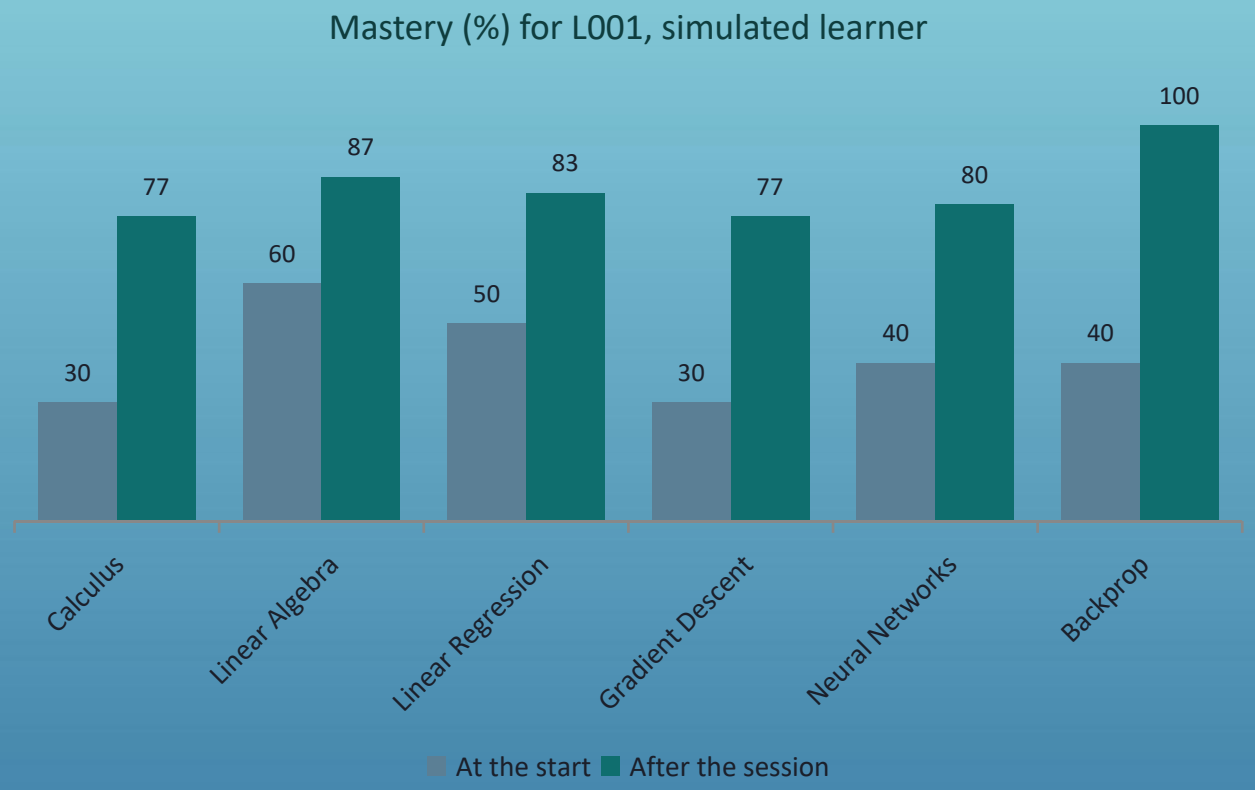
Injected with the "faulty drafts" switch in the Streamlit UI, then fixed automatically on attempt 2.

Results from the evaluation suite

11/11 scenario tests passed

5/5 faulty plans and quizzes caught

288/288 quiz selections valid



Four learners, four behaviours

Ananya: root causes found; Gradient Descent re-taught; all 6 topics cleared

Rahul: one root cause blocks another; both cleared

Sneha: needs 6.5 h, has 4 h, plan says so; Calculus escalated to a tutor

Karan: no gaps; challenge mode, second round, stretch topic cleared

Limitations and future scope

Limitations

- Dummy data and a simulated learner: no real learning gains were measured
- Mock mode uses keyword rules and templates; LLM output was tested using Ollama.
- Mastery is a simple estimate and the question bank is small
- The Critic catches wrong plans and bad questions, not weak teaching

Future scope

- Real course content, a larger question bank, flashcards, and spaced review
- Retrieval over lecture notes, and MCP for the LMS and calendar
- Better mastery models: item response theory, Bayesian knowledge tracing
- A pilot with real learners, LangSmith tracing and an LLM judge